Process control methods for fin-FET gate uniformity at sub-7-nanometre nodes

US9876543 · Macrosilicon Manufacturing Corp.

PSS 78.91

S

4 / 183 • Top 2.2%

STATUS	Active
FILING JURISDICTION	US
OWNER	Macrosilicon Manufacturing Corp.
FAMILY COUNTRIES	US · TW · KR · EP · CN · JP
IPC NUMBER	H01L 21/336
IPC CATEGORY	Semiconductor process — gate fabrication
FILED	2018-06-04
GRANTED	2021-09-14
CLAIMS	32 (4 indep · 28 dep)
REMAINING LIFE	~12 yrs · until 2038-06-04

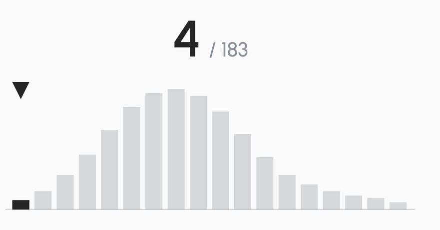
COHORT PERCENTILE

97.8%

183-patent semiconductor advanced-node process pool

RANK IN POOL

4 / 183



Re-ranked weekly as cohort updates

PSS SCORE

78.91

0 50 100

Composite of 8 pillars · 50 indicators

INDICATORS GRADED	50 / 50
COHORT NUMBER	H01L
COHORT NAME	Semiconductor advanced-node process
COHORT SIZE	183 patents
FAMILY JURISDICTIONS	US · TW · KR · EP · CN · JP

SUMMARY

A wafer-level process-control loop that ties **real-time electron-beam metrology** to gate-etch endpoint detection, materially tightening gate-CD uniformity at the 5–7 nm node. Applied during fin-FET fabrication, it cuts cross-wafer CD variance and improves yield in high-mobility-channel architectures.

RATING RATIONALE

Top 2.2% in its industry pool. **Market Coverage (P4 · 86.50)** and **Citation Impact (P1 · 84.20)** are the rating anchors — full six-jurisdiction footprint and a top-decile citation tail in advanced-node process IP. Stable across all five scenarios at 96%+; ranked #4 on the anchor scenario. Forward-Looking Value (P8 · 68.50) is the softest pillar but still well above pool median. Read as **a portfolio anchor of the highest grade** — the kind of asset an IC memo opens with.

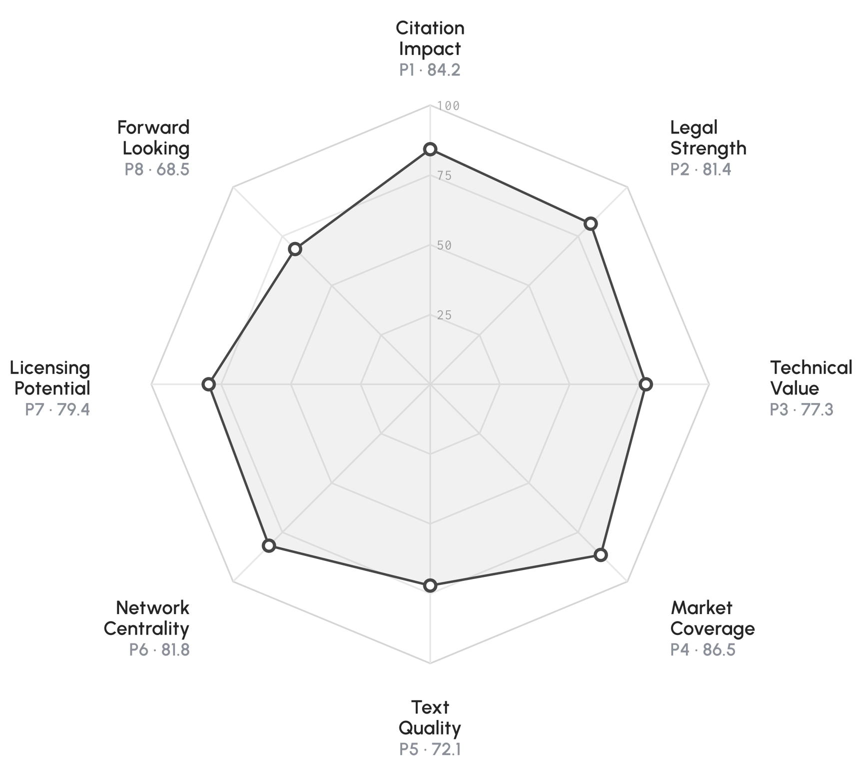
How is this calculated?

Pillars & Indicators

Eight pillars · 50 indicators graded — how the patent strength score is decomposed.

PILLAR ANALYSIS

The Patent Strength Score is a weighted composite of eight pillars, each graded against 183 peers in the cohort.



PILLAR		SCORE	READ
P1	Citation Impact	84.20	KEY DRIVER
P2	Legal Strength	81.40	KEY DRIVER
P3	Technical Value	77.30	SUPPORTIVE
P4	Market Coverage	86.50	KEY DRIVER
P5	Text Quality	72.10	SUPPORTIVE
P6	Network Centrality	81.80	KEY DRIVER
P7	Licensing Potential	79.40	SUPPORTIVE
P8	Forward-Looking Value	68.50	SUPPORTIVE

≥ 70 · TOP QUARTILE 35–70 · MID BAND < 35 · BELOW MEDIAN COHORT MEDIAN (50)

PILLAR DEEP DIVE

Each pillar decomposes into 4–8 individual indicators. Percentiles below are in-pool — derived against the 183-patent semiconductor advanced-node process cohort.

Pillar 1	Pillar 2	Pillar 3	Pillar 4	Pillar 5	Pillar 6	Pillar 7	Pillar 8
Citation Impact 84.20	Legal Strength 81.40	Technical Value 77.30	Market Coverage 86.50	Text Quality 72.10	Network Centrality 81.80	Licensing Potential 79.40	Forward-Looking Value 68.50

CODE	INDICATOR	VALUE	PERCENTILE	WEIGHT
P1.1	Forward citations	47	99.5 %	20%
P1.2	Citation velocity	4.17/yr	98.9 %	20%
P1.3	Generality index	0.87	91.5 %	15%
P1.4	Originality index	0.77	84.2 %	10%
P1.5	Disruption index (CD)	0.24	76.9 %	15%
P1.6	Backward citations	19	69.5 %	10%
P1.7	Examiner-added citation ratio	0.66	62.2 %	5%

Citation Impact is reading **top-quartile**. Forward citations and velocity both top-decile. The drivers concentrated here anchor the rating.

Risk & Action

What anchors the rating, what could move it under stress, and where to focus next.

KEY RATING DRIVERS & SENSITIVITIES

Top-10 indicator readings, ordered by distance from the cohort median. Drivers anchor the rating; sensitivities mark the levers a stress scenario would pull.

KEY RATING DRIVERS — Top 10 positive

01	P4.2	GDP-weighted coverage	98.2 %
02	P1.2	Citation velocity	97.8 %
03	P6.1	PageRank centrality	97.0 %
04	P4.4	Key-market hit rate	96.4 %
05	P1.1	Forward citations	95.6 %
06	P2.7	Continuation / division	93.1 %
07	P3.5	Technical novelty (NLP)	92.5 %
08	P7.6	Choke-point position	91.4 %
09	P5.1	Claim clarity (NLP)	89.8 %
10	P6.2	Betweenness centrality	88.7 %

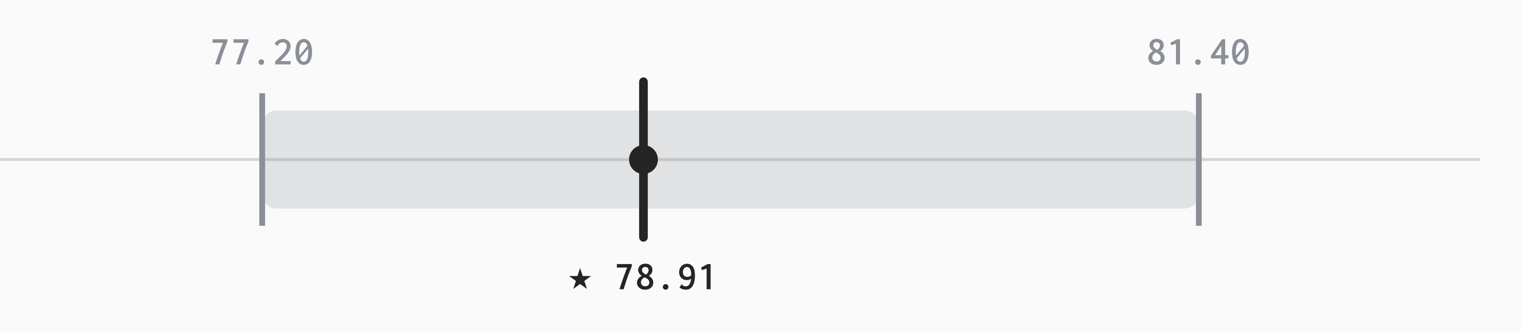
KEY RATING SENSITIVITIES — Top 5 negative

01	P8.5	Future application potential	58.2 %
02	P8.2	Citation-trend slope	61.4 %
03	P5.4	Multi-dep. claim diffusion	62.0 %
04	P3.1	IPC breadth	64.5 %
05	P7.2	Design-around difficulty	66.8 %

SCENARIO SENSITIVITY

Stress-tested under five reweighting stacks — each models a different decision lens. Stability across all five is the strongest single signal of portfolio-anchor quality.

PSS SPREAD ACROSS 5 SCENARIOS — **4.20 pt range** · 77.20 – 81.40 · anchor at **78.91**



SCENARIO	WEIGHT STACK	PSS	COHORT %	RANK	VERDICT
Licensing — Original★ Anchor scenario · default weight stack	Balanced baseline	78.91	97.8 %	#4 / 183	ANCHOR
Licensing — Expert Licensing — Expert framing	P1↑ · P7↑ · P4↑	79.10	97.8 %	#4 / 183	AFFIRMED
Litigation strength Litigation strength framing	P2↑ · P5↑ · P7↑	77.20	96.8 %	#6 / 183	AFFIRMED
Investment / M&A Investment / M&A framing	P1↑ · P6↑ · P8↑	78.45	97.2 %	#5 / 183	AFFIRMED
Tech layout Tech layout framing	P3↑ · P6↑ · P8↑	81.40	97.8 %	#4 / 183	HIGHEST READ

STRESS-TEST OUTCOME

All five scenarios land at **96.8%+** with PSS in a tight **77.20–81.40** band — the rating does not depend on a particular reading. **Tech layout (P3↑ P6↑ P8↑)** gives the highest score at 81.40; **Litigation strength** is the softest at 77.20 but still in the top 4% of the pool. This is one of the pool's anchor-grade assets across every decision lens.

How is this calculated?

ACTION CONSIDERATIONS

RECOMMENDED ACTIONS

01 **License cross-portfolio with foundry customers.** 97.8% cohort percentile and #4 rank make this a defensible anchor for an advanced-node licensing thesis — target TSMC / Samsung / Intel ecosystems.

02 **Lead negotiations with P4 = 86.50.** Six-jurisdiction family is the strongest single bargaining chip — show the geographic moat in opening decks.

03 **Bundle with adjacent gate-CD and lithography filings.** Centrality + family breadth makes this a natural anchor for a multi-patent licensing program.

04 **Cite the Tech layout scenario (81.40)** in the IC memo — highest read across the stress test.

WATCH ITEMS

01 **Forward-Looking Value (P8 · 68.50).** Above median, but the citation slope is flattening as the industry moves to GAA / 3D-stacking architectures.

02 **Future application potential (P8.5 · 58.2%).** Single mid-band reading — pair with a forward-looking GAA filing to backfill the runway thesis.

03 **Design-around difficulty (P7.2 · 66.8%).** Mid-band — competitors may engineer around with alternative endpoint-detection schemes.

04 **Mature-node optionality.** Process IP at this node faces a ~10-year monetization window; plan continuations now.